

A half-exponent bound for the block complexity of algebraic numbers

Abstract

Let ξ be a real algebraic irrational number and $b \geq 2$ an integer. Let $p(L) = p_{\xi,b}(L)$ count the distinct blocks of length L in the fractional base- b expansion of ξ . We prove

$$\limsup_{L \rightarrow \infty} \frac{p(L)(\log \log L)^\gamma}{L\sqrt{\log L}} = +\infty \quad (\gamma > 1).$$

Consequently $p(L)/(L(\log L)^\eta)$ is unbounded for every $\eta < 1/2$. The new input is a sparse-parameter bound for semistable twisted heights in dimension three with fixed forms and fixed finite support, while the weights may vary from one parameter to the next. We prove this bound by adapting the auxiliary-polynomial argument of Evertse and Ferretti. The number of strongly separated parameters is $O(\delta^{-2})$ and the uniform cutoff has double logarithm $O(\delta^{-2} \log(2/\delta))$. An explicit calculation verifies semistability for the full family of weights arising from repetitions. The argument includes nonreal conjugates and composite bases.

1 Introduction

Fix a real algebraic irrational number ξ and an integer $b \geq 2$. Replacing ξ by its fractional part leaves its block complexity unchanged, so we henceforth assume $0 < \xi < 1$. Write $\xi = \sum_{i \geq 1} d_i b^{-i}$, where $d_i \in \{0, \dots, b-1\}$, and put

$$p(L) = p_{\xi,b}(L) = \#\{d_{j+1} \cdots d_{j+L} : j \geq 0\}.$$

We use natural logarithms, and expressions involving iterated logarithms are restricted to sufficiently large arguments.

Theorem 1.1. *For every real algebraic irrational ξ , every integer $b \geq 2$, and every real $\gamma > 1$,*

$$\limsup_{L \rightarrow \infty} \frac{p(L)(\log \log L)^\gamma}{L(\log L)^{1/2}} = +\infty. \tag{1.1}$$

In particular, for every real $\eta < 1/2$,

$$\limsup_{L \rightarrow \infty} \frac{p(L)}{L(\log L)^\eta} = +\infty. \tag{1.2}$$

These are limsup statements. Neither the unweighted endpoint $\eta = 1/2$ nor the case $\gamma = 1$ is asserted. Adamczewski and Bugeaud proved $p_{\alpha,b}(L)/L \rightarrow \infty$ for algebraic irrational α and integer bases $b \geq 2$ [1]. Bugeaud and Evertse obtained the logarithmic limsup refinement with $\eta < 1/11$ [2, Theorem 2.1]. Our argument uses their repetition method and the semistable machinery in [3, Sections 8–14].

The essential issue is the ratio $\rho = r/t$ in an approximation

$$|(b^t - b^r)\xi - a| \leq b^{-v}.$$

A partition of ρ into intervals of length comparable to ε permits fixed weights, but costs a factor of order ε^{-1} . We instead retain the exact value of ρ at each point. This requires a separate argument: the fixed-weight statement of [3, Theorem 8.1] does not itself allow this substitution. Section 5 proves the needed variant, including its cutoff. Its use here depends on an explicit semistability calculation in Section 4.

All constants in the moving-weight argument depend only on the fixed number field, forms, and finite set of places. They are independent of the individual weights and parameters. No effective numerical starting length for (1.1) is claimed. The exponent range is the same for every fixed pair (ξ, b) ; the constants and cutoffs may depend on that pair.

2 Repetitions and a sparse sequence

Fix $u > 0$, $\gamma \geq 0$, and set

$$F(x) = (\log x)^u (\log \log x)^{-\gamma}.$$

This function eventually increases to infinity, satisfies $F(x) = o(x)$, and, for every fixed $D > 0$,

$$F(\lceil Dx F(x) \rceil) / F(x) \rightarrow 1. \quad (2.1)$$

Indeed, its logarithmic derivative is $(u - \gamma / \log \log x) / (x \log x)$; also $\log \lceil Dx F(x) \rceil = \log x + O(\log \log x)$. Suppose, for a contradiction to be specified at the end, that

$$p(k) \leq CkF(k) \quad (k \geq k_0), \quad (2.2)$$

where $C \geq 1$ is fixed and k_0 is sufficiently large.

Lemma 2.1. *Put $a_0 = (24C)^{-1}$. For every sufficiently large integer ℓ there are integers r, s, t, v, a such that*

$$t = r + s \leq \ell, \quad 0 \leq r < t, \quad s \geq v \geq a_0 \ell / F(\ell), \quad 0 \leq a \leq b^t - b^r, \quad (2.3)$$

and

$$|(b^t - b^r)\xi - a| \leq b^{-v}. \quad (2.4)$$

Proof. Take $k = \lfloor \ell / (4CF(\ell)) \rfloor$. For large ℓ , $k \geq k_0$, $k \geq \ell / (8CF(\ell))$, and $p(k) \leq \ell/4$. Among the $\ell - k + 1$ length- k blocks of the prefix there are two equal ones, with starting-position difference D . If $D \geq k$ they are disjoint. If $D < k$, the substring of length $k + D$ has period D , and its first two blocks of length $v' = D \lfloor (k + D) / (2D) \rfloor$ are equal and disjoint. Here $v' \geq k/3$: for $D \geq k/3$ use $v' \geq D$, and otherwise use $v' > (k - D)/2$. Thus in either case the prefix has the form $UVWVX$ with $v = |V| \geq a_0 \ell / F(\ell)$.

Set $r = |U|$, $s = |VW|$, $t = r + s$, $A_j = \sum_{i=1}^j d_i b^{j-i}$, $A_0 = 0$, and $a = A_t - A_r$. The number $a/(b^t - b^r)$ has base- b expansion $U(VW)^\infty$ and agrees with ξ through the first $t + v$ digits. Hence its distance from ξ is at most b^{-t-v} , which gives (2.4). Writing P for the value of VW gives $a = (b^s - 1)A_r + P$, so $0 \leq a \leq b^t - b^r$. The estimate remains valid when the periodic tail consists entirely of the digit $b - 1$, by using the closed interval of values with a specified finite prefix. The length conditions follow from disjointness. $\square$

Fix one such choice for each ℓ .

Lemma 2.2. *There are increasing integers ℓ_j and associated approximations r_j, s_j, t_j, v_j, a_j satisfying*

$$s_{j+1} > 2t_j, \quad t_{j+1} > 2t_j, \quad (2.5)$$

and

$$j \ll \log \ell_j \ll j \log(2j), \quad \log \log \ell_j \asymp \log(2j). \quad (2.6)$$

Proof. Choose ℓ_1 large. Define ℓ_{j+1} as the first $\ell > \ell_j$ with $s_\ell > 2t_j$. It exists since $s_\ell \geq a_0 \ell / F(\ell) \rightarrow \infty$. For fixed $D > 4/a_0$, (2.1) shows that $\ell^* = \lceil D \ell_j F(\ell_j) \rceil$ satisfies $s_{\ell^*} \geq (a_0 D/2) \ell_j > 2t_j$ for large ℓ_j . Thus $\ell_{j+1} \ll \ell_j F(\ell_j)$. With $y_j = \log \ell_j$, this yields $y_{j+1} \leq y_j + u \log y_j + O(1)$, hence $y_j \leq A j \log(2j)$ by induction for a sufficiently large fixed A . For example choose A so that $u \log(A j \log(2j)) + O(1) \leq A \log(2j)$ for all $j \geq 1$. The reverse bound follows from $\ell_j \geq t_j > 2^{j-1} t_1$. Taking logarithms gives the last assertion of (2.6). $\square$

3 Twisted heights and the exact weights

For a number field K write M_K for its places, normalized by $\|z\|_v = |z|_p^{d_v}$ on $z \in \mathbb{Q}$ when $v \mid p$, where $d_v = [K_v : \mathbb{Q}_p]/[K : \mathbb{Q}]$ and $\sum_{v \mid p} d_v = 1$. At an infinite place represented by an embedding σ_v the corresponding formula is $\|z\|_v = |\sigma_v(z)|^{d_v}$. For a system of linearly independent forms $\mathcal{L} = (L_{iv} : v \in M_K, 1 \leq i \leq 3)$ and real weights c_{iv} , put

$$H_{\mathcal{L}, c, Q}(x) = \prod_{v \in M_K} \max_{1 \leq i \leq 3} (\|L_{iv}(x)\|_v Q^{-c_{iv}}). \quad (3.1)$$

For $x \in \overline{\mathbb{Q}}^3$ use a finite extension E/K containing its coordinates, keep the forms unchanged at $w \mid v$, and multiply the weights by $d(w \mid v) = [E_w : K_v]/[E : K]$. This gives an extension-independent height.

Choose a finite Galois extension $K/\mathbb{Q}$ in $\mathbb{C}$ containing ξ , and let $G = \text{Gal}(K/\mathbb{Q})$ and $d = [\mathbb{Q}(\xi) : \mathbb{Q}] \geq 2$. Write $\beta_1, \dots, \beta_d$ for the distinct conjugates of ξ . They are all nonzero. For each infinite place v , choose $\sigma_v \in G$ with $\|z\|_v = |\sigma_v(z)|^{d_v}$, and set $\beta_v = \sigma_v^{-1}(\xi)$. Define the fixed forms by

$$(L_{1v}, L_{2v}, L_{3v}) = \begin{cases} (X_1, X_2, L_{\beta_v}), & v \mid \infty, \\ (X_1, X_2, X_3), & v \nmid \infty, \end{cases} \quad L_\beta = -\beta X_1 + \beta X_2 + X_3. \quad (3.2)$$

Their local determinants are all 1, and there are $d+3$ distinct forms. For every rational vector x ,

$$\|L_{\beta_v}(x)\|_v = |\sigma_v(L_{\beta_v}(x))|^{d_v} = |L_\xi(x)|^{d_v}. \quad (3.3)$$

This is the inverse-conjugate transport used in [2, Section 5, equations (5.14)–(5.17)]. It uses smallness only at the specified real value of ξ .

Lemma 3.1 (Mass of a conjugate). *For every conjugate β of ξ ,*

$$\sum_{\substack{v \mid \infty \\ \beta_v = \beta}} d_v = \frac{1}{d}. \quad (3.4)$$

Proof. Let $\iota \in G$ be complex conjugation on K and $H = \langle \iota \rangle$. Infinite places are represented by the left cosets $H\sigma$. Since ξ is real, $(\iota\sigma)^{-1}(\xi) = \sigma^{-1}(\xi)$, so β_v does not depend on the representative. The field K is either totally real or totally imaginary, and $d_v = |H|/|G|$ in either case. The set of $\sigma \in G$ with $\sigma^{-1}(\xi) = \beta$ has cardinality $|G|/d$ by orbit–stabilizer. It is a union of left H -cosets, each of cardinality $|H|$. Its corresponding total place weight is therefore $(|G|/(d|H|))(|H|/|G|) = 1/d$. $\square$

For each rational prime $p \mid b$, put

$$\omega_p = \frac{\text{ord}_p(b) \log p}{\log b}, \quad \kappa_v = d_v \omega_p \quad (v \mid p). \quad (3.5)$$

These numbers are positive and satisfy $\sum_{p \mid b} \omega_p = 1$ and $\sum_{v \mid b} \kappa_v = 1$, where $v \mid b$ means $v \mid p$ for some $p \mid b$. Let S consist of all infinite places and all places above the prime divisors of b .

Fix $0 < \varepsilon \leq 1/2$ and $0 \leq \rho \leq 1 - \varepsilon$. In the order of (3.2), define raw weights by

$$e_v = \begin{cases} d_v(1, \rho, -\varepsilon), & v \mid \infty, \\ \kappa_v(-1, -\rho, 0), & v \mid b, \\ (0, 0, 0), & v \notin S. \end{cases} \quad (3.6)$$

Put

$$\bar{e}_v = \frac{1}{3} \sum_i e_{iv}, \quad D_\varepsilon = 1 + \varepsilon/3, \quad c_{iv}(\rho) = \frac{e_{iv} - \bar{e}_v}{D_\varepsilon}, \quad \delta = \frac{\varepsilon}{3 + \varepsilon}. \quad (3.7)$$

The two mass identities give

$$\sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} = 1, \quad \sum_v \bar{e}_v = -\varepsilon/3. \quad (3.8)$$

Indeed $\sum_{v,i} e_{iv} = -\varepsilon$ and the sum of the raw local maxima is 1; centering changes the latter to $1 + \varepsilon/3$.

For a fixed integer J , take $\varepsilon = a_0/F(\ell_J)$ and $j \leq J$. By monotonicity of F and (2.3), $v_j \geq \varepsilon t_j$ and $t_j - r_j \geq v_j$, so $\rho_j = r_j/t_j \leq 1 - \varepsilon$. Set $x_j = (b^{t_j}, b^{r_j}, a_j)$, $\Psi_j = b^{t_j}$ and $Q_j = \Psi_j^{D_\varepsilon}$. At the infinite places, (2.4) and (3.3) give $\|L_{iv}(x_j)\|_v \leq \Psi_j^{e_{iv}(\rho_j)}$. At a place $v \mid p \mid b$, the same bounds follow from

$$\|b^{t_j}\|_v = \Psi_j^{-\kappa_v}, \quad \|b^{r_j}\|_v = \Psi_j^{-\kappa_v \rho_j}, \quad \|a_j\|_v \leq 1.$$

At all other finite places the coordinates are integral. Hence

$$H_{\mathcal{L}, c(\rho_j), Q_j}(x_j) \leq \Psi_j^{\sum_v \bar{e}_v} = Q_j^{-\delta}, \quad Q_{j+1} > Q_j^2. \quad (3.9)$$

No primitivity of x_j is needed; Ψ_j is an external parameter, not an identification with its projective height.

4 Uniform semistability

For a q -dimensional subspace $U \subset \overline{\mathbb{Q}}^3$, define its local weight as the minimum of $\sum_{i \in I} c_{iv}$ over all q -element sets I for which $(L_{iv}|_U)_{i \in I}$ is linearly independent. Its global weight $w_c(U)$ is the sum over places. The system is semistable if $w_c(U) \leq 0$ for every nonzero proper U . Define w_e analogously. Centering gives

$$w_c(U) = \frac{w_e(U) + q\varepsilon/3}{D_\varepsilon}. \quad (4.1)$$

Proposition 4.1. *For the family (3.6)–(3.7), all nonzero proper subspaces have negative weight. More precisely,*

$$w_c(U) \leq -\frac{\varepsilon}{6D_\varepsilon} \quad (\dim U = 1), \quad w_c(U) \leq -\frac{\varepsilon}{3D_\varepsilon} \quad (\dim U = 2).$$

Proof. By Lemma 3.1, the infinite-place contribution to $w_e(U)$ can be computed using each distinct L_β once, with raw weights $(1, \rho, -\varepsilon)/d$. The finite contributions combine into one minimum calculation with coordinate forms and weights $(-1, -\rho, 0)$, since $\sum_{v|b} \kappa_v = 1$. This grouping concerns only the subspace weight calculation; all original places and norms remain in the height.

Consider a line U . If no L_β vanishes on it, its infinite contribution is $-\varepsilon$ and its finite contribution is at most zero. If two distinct L_β vanish, subtracting them shows $U = \langle (1, 1, 0) \rangle$; all L_β then vanish and $w_e(U) = \rho - 1 \leq -\varepsilon$. Otherwise precisely one L_β vanishes. If $X_1|_U \neq 0$, the exceptional infinite contribution is at most $1/d$, the remaining infinite contribution is $-(1 - 1/d)\varepsilon$, and the finite contribution is -1 . Thus $w_e(U) \leq -(1 - 1/d)(1 + \varepsilon)$. If $X_1|_U = 0$, then $X_2|_U \neq 0$ and $w_e(U) = -(1 - 1/d)(\rho + \varepsilon)$. Since $d \geq 2$, every line therefore has $w_e(U) \leq -\varepsilon/2$.

For planes write $U = \ker M$, and put $A_\beta = \text{span}(L_\beta, X_2)$ and $B = \text{span}(X_1, X_2)$ in the dual space. For distinct conjugates β, β' ,

$$A_\beta \cap A_{\beta'} = A_\beta \cap B = \langle X_2 \rangle.$$

This uses $\beta \neq \beta'$ and $\beta \neq 0$. At a grouped infinite place the minimum pair has weight $(\rho - \varepsilon)/d$ unless $M \in A_\beta$; the finite minimum is $-1 - \rho$ unless $M \in B$. The following list is exhaustive. Proportional normals are identified.

Position of M	$w_e(\ker M)$
$M \notin B \cup \bigcup_\beta A_\beta$	$-1 - \varepsilon$
$M \in A_\beta \setminus (\langle L_\beta \rangle \cup \langle X_2 \rangle)$	$-1 - \varepsilon + (1 - \rho)/d$
$M \in \langle L_\beta \rangle$ for some β	$-(1 - 1/d)(1 + \varepsilon)$
$M \in B \setminus (\langle X_1 \rangle \cup \langle X_2 \rangle)$	$\rho - 1 - \varepsilon$
$M \in \langle X_1 \rangle \cup \langle X_2 \rangle$	$-\varepsilon$

Each entry is at most $-\varepsilon$, since $d \geq 2$, $\varepsilon \leq 1/2$ and $\rho \leq 1 - \varepsilon$. For example, at $M = L_\beta$ the pair for that conjugate must be (X_1, X_2) , whereas at $M = X_1$ the finite pair is (X_2, X_3) . These observations also cover the ties when $\rho = 0$. Equation (4.1) proves both bounds. $\square$

5 A finite-support moving-weight bound

We prove the arithmetic input separately. In this section K , S , and $\mathcal{L}$ are fixed; the dimension is three. Assume S is finite and contains all infinite places, $\mathcal{L}$ is the coordinate system outside S , and the determinant at every place is 1. Write $s = |S|$. Let $v_0 \notin S$ be a fixed finite place. Weights called admissible below satisfy

$$c_{iv} = 0 \ (v \notin S), \quad \sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} \leq 1, \quad w_c(U) \leq 0 \ (0 \neq U \subsetneq \overline{\mathbb{Q}}^3). \quad (5.1)$$

Theorem 5.1. *There are constants depending only on the fixed data with the following property. For $0 < \delta \leq 1$ there exist $\Omega(\delta) > 2$ and $C_*(\delta) > e$ with*

$$\Omega(\delta) \ll \delta^{-5}, \quad \log \log C_*(\delta) \ll \delta^{-2} \log(2/\delta), \quad (5.2)$$

such that any sequence $(c^{(h)}, Q_h)$ of admissible weights and parameters with

$$Q_1 \geq C_*(\delta), \quad Q_{h+1} > Q_h^{\Omega(\delta)}, \quad \{x \neq 0 : H_{\mathcal{L}, c^{(h)}, Q_h}(x) \leq Q_h^{-\delta}\} \neq \emptyset$$

has length $O(\delta^{-2})$.

We first isolate the pointwise part of the proof in [3]. This is needed to control the cutoff, rather than import the larger constant in their final interval theorem.

Lemma 5.2 (Pointwise exterior data). *There are constants $A_0 > 0$ and $c_0 > 0$, depending only on $K, \mathcal{L}$, such that if c is admissible, $Q \geq \exp(A_0/\delta)$ and $H_{\mathcal{L},c,Q}(x) \leq Q^{-\delta}$ for some nonzero x , then there is $k \in \{1, 2\}$ with the following properties. Put $N = \binom{3}{k} = 3$. There are fixed exterior forms $\hat{L}_{lv}$, depending only on $k, \mathcal{L}$, and independent vectors $z_1, z_2 \in \overline{\mathbb{Q}}^3$ spanning a plane T such that*

$$H_2(T) \geq Q^{c_0\delta}. \quad (5.3)$$

For every place w of a common finite extension E/K ,

$$\|\hat{L}_{lw}(z_j)\|_w \leq \begin{cases} Q^{d(w|v)a_{lv}}, & w \mid v \neq v_0, \\ Q^{d(w|v_0)b_l}, & w \mid v_0, \end{cases} \quad (5.4)$$

where, with $\gamma_v = \max_i c_{iv}$,

$$\sum_l a_{lv} = 0, \quad |a_{lv}| \leq 2\gamma_v, \quad a_{lv} = 0 \ (v \notin S), \quad \sum_l b_l \leq -\delta/3, \quad |b_l| \leq 3. \quad (5.5)$$

Here H_2 is the absolute Euclidean height of the exterior coordinate vector defining a subspace, as in [3, Section 6].

Proof. Let $\lambda_1 \leq \lambda_2 \leq \lambda_3$ be the successive infima of the twisted height over $\overline{\mathbb{Q}}$. The fixed-form comparison and the absolute Minkowski theorem [3, Lemma 7.1 and Proposition 9.2] give

$$C_0^{-1}Q^{-1} \leq \lambda_1, \quad 3^{-3/2} \leq \lambda_1\lambda_2\lambda_3 \leq 8. \quad (5.6)$$

The product bounds use the determinant assumption. As $\lambda_1 \leq Q^{-\delta}$, enlarging A_0 ensures $\lambda_3 \geq 1$. One of the two consecutive ratios is therefore at most $Q^{-\delta/2}$; fix such a k . Its infimum subspace T_k has dimension k .

For clarity we also give the height estimate behind this assertion. Choose a basis $g_1, \dots, g_k$ with twisted heights arbitrarily close to $\lambda_1, \dots, \lambda_k$, and at each place select an independent restricted k -tuple realizing its local minimum weight. Semistability and the determinant estimate imply that the product of these local determinant norms is at most $C\lambda_1 \cdots \lambda_k \leq C'Q^{-\delta/3}$. The last inequality follows from (5.6) and the gap, since $k(3-k)/6 = 1/3$ for $k = 1, 2$. Let $R \geq 3$ bound the number of distinct original forms. Their nonzero k -fold determinants on the basis number at most $\binom{R}{k}$. The product formula, applied to these determinants, gives the lower bound $(C''H_2(T_k))^{-((\binom{R}{k})-1)}$ for the same product; this is also [3, Lemma 10.2]. The fixed constants can be absorbed when $\log Q \geq A_0/\delta$, giving $H_2(T_k) \geq Q^{c_0\delta}$, for example with $c_0 = 1/(12R^3)$ after increasing A_0 .

Apply Davenport's construction [3, Lemma 11.3] separately to this height and take the $(3-k)$ th exterior power as in its equations (11.10)–(11.17). The exterior forms depend only on $k, \mathcal{L}$. Away from v_0 the exponents a_{lv} are sums of $3-k$ original weights; their sums are zero and $|a_{lv}| \leq 2\gamma_v$. The first $N-1 = 2$ exterior vectors span T , whose height equals $H_2(T_k)$ by [3, Lemma 6.1].

Here is the size check at the auxiliary place, including the cutoff. Let ν_l be the products of $3-k$ distinct successive infima. Up to a permutation, the bounds defining Q^{b_l} are $3^{27}\nu_1, \dots, 3^{27}\nu_{N-1}, 3^{27}\nu_{N-1}$. Thus their product is

$$3^{27N}(\lambda_1\lambda_2\lambda_3)^{\binom{2}{2-k}} \frac{\lambda_k}{\lambda_{k+1}} \leq CQ^{-\delta/2} \leq Q^{-\delta/3}.$$

Also (5.6) bounds any product of $p = 1, 2$ distinct infima between $C^{-1}Q^{-p}$ and CQ^{3-p} . It follows that $|b_l| \leq 3$ once A_0 is sufficiently large. All absorptions above have the form $Q^{a\delta} \geq C$ with fixed $a, C > 0$, or $Q^a \geq C$. Consequently $\exp(A_0/\delta)$ suffices throughout; the final cutoff of [3, Theorem 8.1] has not been used. $\square$

5.1 The auxiliary polynomial with changing weights

Suppose m of the parameters have the same k in Lemma 5.2; label them $h = 1, \dots, m$. Their exterior forms are consequently the same. Write a_{hlv}, b_{hl} for their exterior weights and set

$$\Gamma_v = \max_{1 \leq h \leq m} \max_i c_{iv}^{(h)}. \quad (5.7)$$

Each $\Gamma_v \geq 0$, and admissibility gives

$$\sum_{v \in S} \Gamma_v \leq s. \quad (5.8)$$

All ordinary-place vanishing conditions below are imposed only at places with $\Gamma_v > 0$.

Let $r_1, \dots, r_m$ be positive integers, $D_r = \sum_h r_h$, and let $\mathcal{U}(r)$ consist of arrays $j = (j_{hl})$ of nonnegative integers with $\sum_l j_{hl} = r_h$ for each h . Put $V = |\mathcal{U}(r)|$. For $0 < \alpha < 1$ satisfying

$$m > 2\alpha^{-2} \log(2s + 2), \quad (5.9)$$

there is a nonzero multihomogeneous polynomial P over K , of block degrees r_h , with the following properties. Expand P in the exterior forms at a place v , writing its coefficients as $d_j^{(v)}$. Then

$$d_j^{(v)} = 0 \quad \text{if } \sum_{h,l} j_{hl} a_{hlv} / r_h \geq 3m\alpha\Gamma_v, \quad v \in S, \Gamma_v > 0, \quad (5.10)$$

$$d_j^{(v_0)} = 0 \quad \text{if } \sum_{h,l} j_{hl} b_{hl} / r_h \geq -m\delta/9 + 3m\alpha, \quad (5.11)$$

$$H_2(P) \leq C_K A^{D_r}. \quad (5.12)$$

Here and below $A \geq 2$ and $C_K \geq 1$ are fixed constants independent of $m, r, c^{(h)}, Q_h$.

To prove this, choose j uniformly from $\mathcal{U}(r)$. The blocks are independent, and $r_h^{-1} \sum_l j_{hl} a_{hlv}$ has mean zero and lies in $[-3\Gamma_v, 3\Gamma_v]$. Hoeffding's inequality therefore bounds the fraction satisfying the condition in (5.10) by $e^{-m\alpha^2/2}$. For the special place the mean of the total is at most $-m\delta/9$ and the individual summands lie in $[-3, 3]$, giving the same bound. This is precisely the block-dependent form of [3, Lemma 13.3]. The total number of imposed linear equations in the V coefficients of P is less than $V/2$ by (5.9). The changes of coordinates use fixed matrices. Consequently each row of this linear system has height at most $A_1^{D_r}$ and $V \leq (eN)^{D_r}$, uniformly in which equations were selected. The number-field Siegel lemma [3, Lemma 13.1], with exponent $U/(V - U) < 1$, gives (5.12) after increasing A . This proves all three assertions, without discretizing places or weights.

For a derivative multi-index i , use divided derivatives $P_i = (\prod_{h,l} i_{hl}!)^{-1} \partial^i P$ and write

$$|i|_r = \sum_{h,l} i_{hl} / r_h.$$

If $|i|_r \leq 2m\alpha$, expansion in the exterior forms gives

$$P_i = \sum_j d_{i,j}^{(v)} \prod_{h,l} \widehat{L}_{lv}(X_h)^{j_{hl}}, \quad (5.13)$$

$$d_{i,j}^{(v)} = 0 \quad \text{if } \sum_{h,l} j_{hl} a_{hlv} / r_h > 12m\alpha\Gamma_v, \quad v \in S, \quad (5.14)$$

$$d_{i,j}^{(v_0)} = 0 \quad \text{if } \sum_{h,l} j_{hl} b_{hl} / r_h > -m\delta/9 + 12m\alpha, \quad (5.15)$$

$$\prod_v \max_j \|d_{i,j}^{(v)}\|_v \leq C_K A_2^{D_r}. \quad (5.16)$$

Indeed a contributing original monomial has index $j + q$, where $\sum_l q_{hl} = \sum_l i_{hl}$ in every block. Its weighted shift has absolute value at most $6m\alpha\Gamma_v$, or $6m\alpha$ at v_0 . Either strict inequality above would therefore force that original coefficient to vanish by (5.10) or (5.11). The coefficient bound follows from fixed inverse coordinate matrices and binomial coefficient bounds, exactly as in [3, Lemma 13.4]; combined with (5.12) these give a fixed A_2 . At places outside $S \cup \{v_0\}$ all the weights are zero, so no vanishing condition is required.

5.2 Nonvanishing, the product formula, and the cutoff

We now prove Theorem 5.1. Choose

$$\alpha = \frac{\delta}{1000(s+1)}, \quad m = \left\lfloor 2\alpha^{-2} \log(2s+2) \right\rfloor + 1, \quad \Omega = \frac{4m^2}{\alpha}. \quad (5.17)$$

There are only two possible k , so a sequence of at least $2m - 1$ parameters has a subsequence of m parameters with the same k . Separation is preserved. Suppose such a subsequence exists and label it by $Q_1 < \dots < Q_m$.

Choose an arbitrarily large positive integer r_1 with $r_1 \log Q_1 > \alpha^{-1} \log Q_m$ and $C_K \leq 2^{r_1}$, and put

$$r_h = \left\lceil \frac{r_1 \log Q_1}{\log Q_h} \right\rceil \quad (h > 1), \quad B = Q_1^{r_1}.$$

Then

$$B \leq Q_h^{r_h} < B^{1+\alpha}, \quad r_h \leq r_1, \quad \frac{r_h}{r_{h+1}} > \frac{\Omega}{1+\alpha} \geq \frac{2m^2}{\alpha}. \quad (5.18)$$

In particular the decreasing degree ratio has the direction required by [3, Proposition 12.1, equation (12.3)]. Construct P as above. For each exterior plane T_h , use the grid

$$\mathcal{G}_h = \{u_1 z_{h1} + u_2 z_{h2} : u_j \in \mathbb{Z}, |u_j| \leq \lceil N/\alpha \rceil\}, \quad N = 3.$$

Put $E_m = (N-1)(3m^2/\alpha)^m$ and $A_3 = 2eA$. By (5.12) and $C_K \leq 2^{r_1}$, $e^{D_r} H_2(P) \leq A_3^{D_r}$. The other hypothesis of that nonvanishing proposition is

$$H_2(T_h)^{r_h} \geq (e^{D_r} H_2(P))^{E_m}. \quad (5.19)$$

By (5.3) and (5.18) it is enough that

$$c_0 \delta \log Q_1 \geq m E_m \log A_3. \quad (5.20)$$

The proposition then supplies $x_h \in \mathcal{G}_h \setminus \{0\}$ and $|i|_r \leq 2m\alpha$ such that $P_i(x_1, \dots, x_m) \neq 0$.

We estimate this value. For an infinite place w , let $s(w)$ be its local degree divided by $[E : \mathbb{Q}]$, and set $s(w) = 0$ at finite places. The grid and (5.4) give the extra factor

$$\|\widehat{L}_{lw}(x_h)\|_w \leq (N^2/\alpha)^{s(w)} Q_h^{d(w|v)a_{hlv}} \quad (w \nmid v_0), \quad (5.21)$$

with the same bound at $w \mid v_0$ using b_{hl} and no grid factor. The factor N^2/α is retained throughout.

For a surviving monomial at $w \mid v \neq v_0$, put $A_{hw} = r_h^{-1} \sum_l j_{hl} d(w \mid v) a_{hlv}$ and $\Gamma_w = d(w \mid v) \Gamma_v$. The derivative degrees are at most r_h , so $|A_{hw}| \leq 3\Gamma_w$. Replacing $Q_h^{r_h}$ by B costs at most $3m\alpha\Gamma_w$ in its exponent, by (5.18). Equation (5.14) bounds the remaining sum by $12m\alpha\Gamma_w$. The resulting exponent is at most $15m\alpha\Gamma_w$. At $w \mid v_0$ the same argument using (5.15) gives at most $d(w \mid v_0)(-m\delta/9 + 15m\alpha)$.

The number of monomials in a derivative is at most $V \leq (eN)^{D_r}$. Multiplying the local bounds, using (5.16), $\sum_w s(w) = 1$, $\sum_{w|v} d(w|v) = 1$, and (5.8), gives

$$\begin{aligned} \prod_w \|P_i(x_1, \dots, x_m)\|_w &\leq G(\alpha)^{D_r} (B^m)^{-\delta/9+15(s+1)\alpha} \\ &\leq G(\alpha)^{D_r} (B^m)^{-\delta/12}, \end{aligned} \quad (5.22)$$

where $G(\alpha) = 2A_2eN^3/\alpha$. We used $C_K \leq 2^{r_1} \leq 2^{D_r}$. Since $D_r \leq mr_1$, the right side is strictly less than 1 if

$$\log Q_1 \geq 24\delta^{-1} \log G(\alpha). \quad (5.23)$$

This contradicts the product formula for the nonzero value of P_i .

For definiteness, all the required conditions hold with

$$C_*(\delta) = \exp \left(\frac{A_0 + 1}{\delta} + \frac{2mE_m \log A_3}{c_0\delta} + \frac{24 \log G(\alpha)}{\delta} \right). \quad (5.24)$$

As $m \ll \delta^{-2}$ and $\alpha \asymp \delta$, $\Omega \ll \delta^{-5}$ and

$$\log \log C_*(\delta) \ll m \log(m/\alpha) + \log(2/\delta) \ll \delta^{-2} \log(2/\delta).$$

The maximum length is at most $2m - 2$. This proves the theorem.

Corollary 5.3. *Under the same hypotheses, suppose $Q_{h+1} > Q_h^2$. The number of parameters at or above $C_*(\delta)$ is $O(\delta^{-2} \log(2/\delta))$.*

Proof. Set $q = \lceil \log_2 \Omega \rceil \ll \log(2/\delta)$ and partition the indices modulo q . In each part consecutive parameters satisfy the separation condition of Theorem 5.1. Apply its bound to each part. $\square$

6 Completion of the complexity argument

Apply the moving-weight theorem with the fixed system (3.2) and the finite set S from Section 3. Choose a fixed rational prime $p_0 \nmid b$ and a place above it as v_0 . All admissibility conditions follow from (3.8) and Proposition 4.1. For a prefix sequence of length J constructed under (2.2), use the common $\varepsilon = a_0/F(\ell_J)$ and $\delta = \varepsilon/(3 + \varepsilon)$. These depend on J , whereas the forms, field, and support do not. Hence every implied constant in the arithmetic bound is uniform in J , although it may depend on ξ and b .

Let $y_J = \log \ell_J$. By (2.6),

$$\delta^{-2} \log(2/\delta) \ll F(\ell_J)^2 \log(2F(\ell_J)) \ll J^{2u} (\log(2J))^{2u+1-2\gamma}. \quad (6.1)$$

For the last step use both $J \ll y_J \ll J \log(2J)$ and $\log y_J \asymp \log(2J)$, so that negative powers of $\log y_J$ are also controlled. If $u < 1/2$ and $\gamma = 0$, the last expression is $o(J)$. If $u = 1/2$ and $\gamma > 1$, it is $J(\log(2J))^{2-2\gamma} = o(J)$ as well.

In either case (5.2) implies $\log \log C_*(\delta) = o(J)$. But (2.5) and $Q_j = b^{D_\varepsilon t_j}$ imply

$$\log \log Q_j = \log t_j + O(1) \geq (j-1) \log 2 + O(1).$$

Consequently every j with $\lceil J/2 \rceil \leq j \leq J$ has $Q_j \geq C_*(\delta)$ for sufficiently large J . Corollary 5.3, together with (3.9), now gives

$$J/2 \ll \delta^{-2} \log(2/\delta) = o(J),$$

a contradiction. Taking $u = 1/2$ proves (1.1), since a finite limsup would imply (2.2) for some C . For $0 < \eta < 1/2$ the direct argument with $u = \eta$, $\gamma = 0$ proves (1.2). Alternatively (1.1) implies it for every real $\eta < 1/2$, because $(\log L)^{1/2-\eta}/(\log \log L)^\gamma \rightarrow \infty$. This completes the proof of Theorem 1.1.

7 Scope of the argument

The improvement uses two distinct facts: the finite common support permits simultaneous coefficient vanishing with $m = O(\delta^{-2})$, and the entire family of exact weights is semistable. The first does not follow just by applying a fixed-weight theorem at each point. The second is a calculation about every algebraic subspace, rather than only the rational approximating vectors.

The proof deliberately uses the cutoff in (5.24). Using the cutoff from the published final interval theorem without rechecking its preliminary lemmas would introduce an avoidable additional logarithm in the double-logarithmic threshold estimate. Similarly, the grid factor in (5.21) is harmless only after its contribution is included in that cutoff.

Lemma 3.1 and the normalization $\sum_{v|b} \kappa_v = 1$ make the semistability argument applicable to every real algebraic irrational number and every integer base $b \geq 2$. Nonreal conjugates and splitting or ramification of primes cause no additional loss in the exponent. The finite support, and consequently the constants, are allowed to depend on the fixed pair (ξ, b) . No assertion uniform in a varying base or varying algebraic number is made about these constants.

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